Statistical Signal Processing
Lecture 05:
Kalman filter state (signal) estimation

Literature and Problems: See course schedule

Agenda (preliminary):

Problem setup

- **general block diagram**
- **signal processing oriented application: signal estimation**
- **control system oriented application: plant state estimation**

Kalman filter vs. optimal Wiener filter

Types of Kalman filters

- **CKF/DKF, scalar KF, vector KF**

Proces (signal) model

- **states, matrices, process noise/uncertainty, zero-mean, covariance matrix Q_k**

Observation (measurement) model

- **states, matrices, measurement noise, zero-mean, covariance matrix R_k**

Notation and task for Kalman filter

- $\hat{\mathbf{x}}_{\mathbf{k}}^{-}$, $\hat{\mathbf{x}}_{\mathbf{k}}$, $\hat{\mathbf{z}}_{\mathbf{k}}^{-}$, $\mathbf{P}_{\mathbf{k}}^{-}$, $\mathbf{P}_{\mathbf{k}}$
- **Find best linear filter** $\hat{\mathbf{x}}_{\mathbf{k}} = \mathbf{f}(\hat{\mathbf{x}}_{\mathbf{k-1}}, \mathbf{z}_{\mathbf{k}})$ **minimizing** $\mathbf{P}_{\mathbf{k}}$

Time line for "current" Kalman filter: predict step + update step

Predict step equations: $\hat{\mathbf{x}}_{\mathbf{k}}^{-}$, $\hat{\mathbf{z}}_{\mathbf{k}}^{-}$, $\mathbf{P}_{\mathbf{k}}^{-}$

Update step equations: $\hat{\mathbf{x}}_{\mathbf{k}}$, $\mathbf{K}_{\mathbf{k}}$, $\mathbf{P}_{\mathbf{k}}$

Kalman filter algorithm block diagram including initialization

Kalman filter interpreted as a recursive IIR-filter

Steady state Kalman filter and Ricatti equation

Demos

- **stationary/non-stationary scalar state KF**
- **vector state KF applied for 1D tracking**